Markets

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Chapter 4: Consumer and Producer Surplus

Evaluating the Market Equilibrium

Positive Analysis

Describe the world as it is

Equilibrium price & quantity in a competitive market

Normative Analysis

About how the world should be

Is the market equilibrium outcome "good" or "bad"?

Criteria?

Consumer Surplus

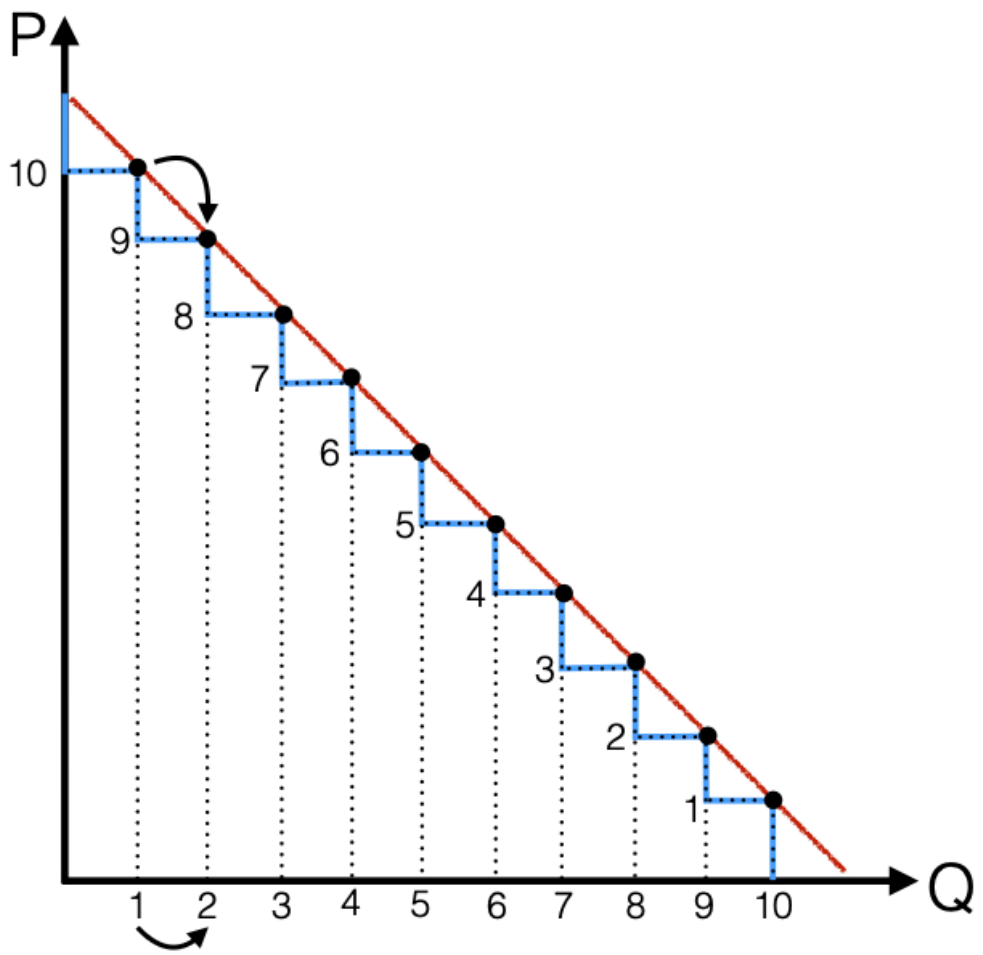
Willingness to Pay

The maximum price that a consumer is willing and able to pay for a good.

It measures how much the consumer values the good or service.

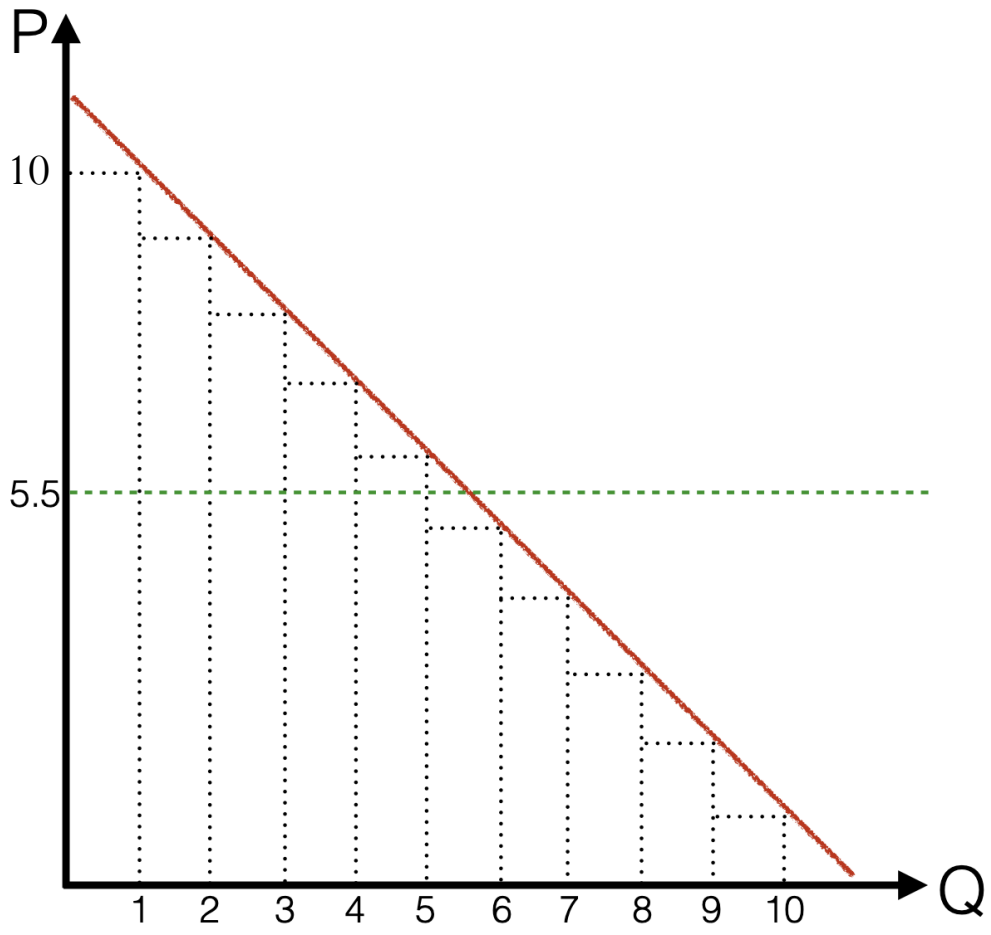
WTP, Demand, and Marginal Benefits

Buyer	WTP	TB	MB
1st	10	10	10
2nd	9	19	9
3rd	8	27	8
4th	7	34	7
5th	6	40	6
6th	5	45	5
7th	4	49	4
8th	3	52	3
9th	2	54	2
10th	1	55	1



Introducing Prices

Buyer	WTP	Price	CS
1st	10	5.5	4.5
2nd	9	5.5	3.5
3rd	8	5.5	2.5
4th	7	5.5	1.5
5th	6	5.5	0.5
6th	5	5.5	
7th	4	5.5	
8th	3	5.5	
9th	2	5.5	
10th	1	5.5	



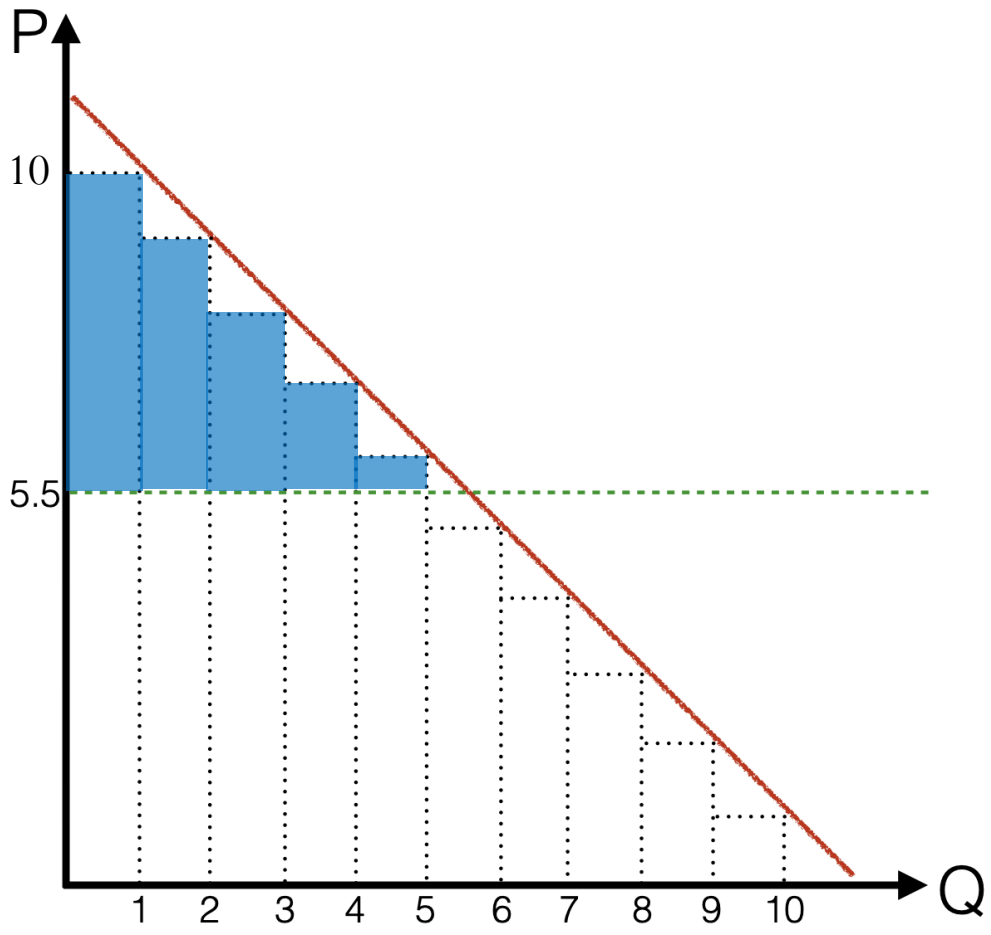
Consumer Surplus

The amount a buyer is willing to pay for a good minus the amount she actually needs to pay for it.

It measures the benefit that buyers receive from a good as they themselves perceive it.

Introducing Prices

Buyer	WTP	Price	CS
1st	10	5.5	4.5
2nd	9	5.5	3.5
3rd	8	5.5	2.5
4th	7	5.5	1.5
5th	6	5.5	0.5
6th	5	5.5	0
7th	4	5.5	0
8th	3	5.5	0
9th	2	5.5	0
10th	1	5.5	0



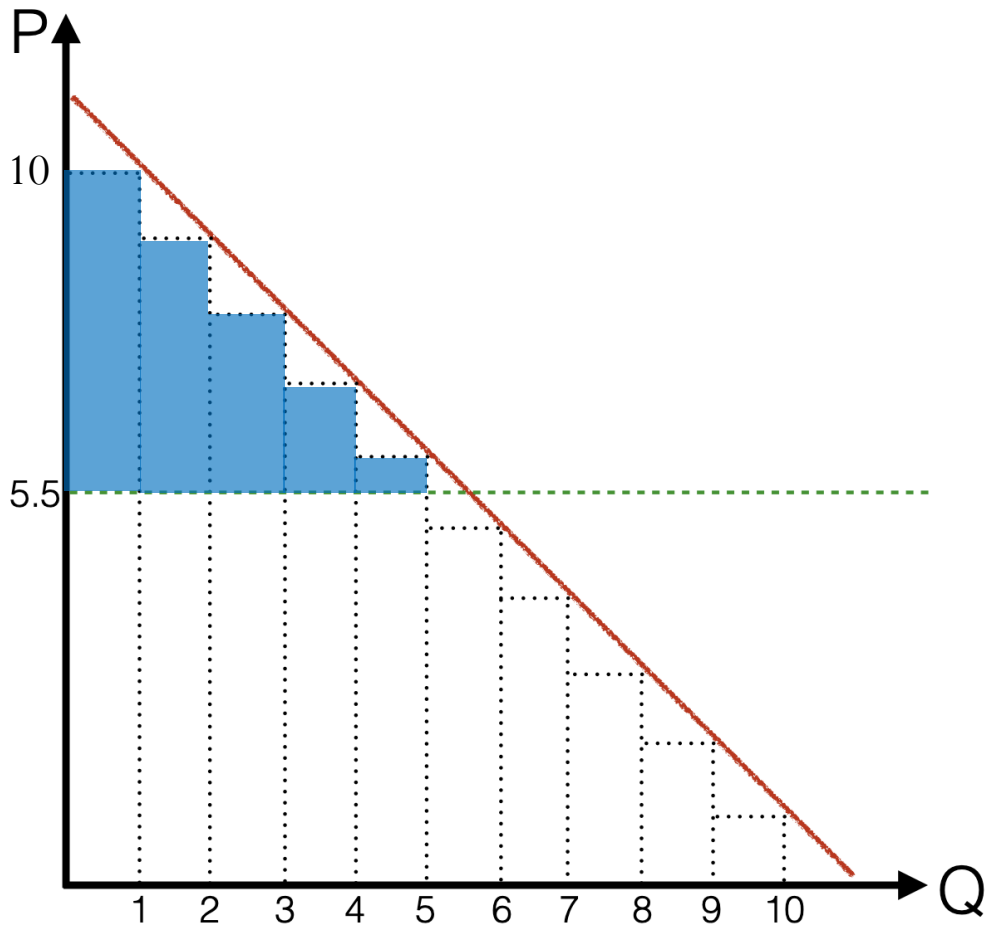
Introducing Prices

Marginal Buyer's Surplus

For any given quantity, the vertical distance b/w the demand curve between price line.

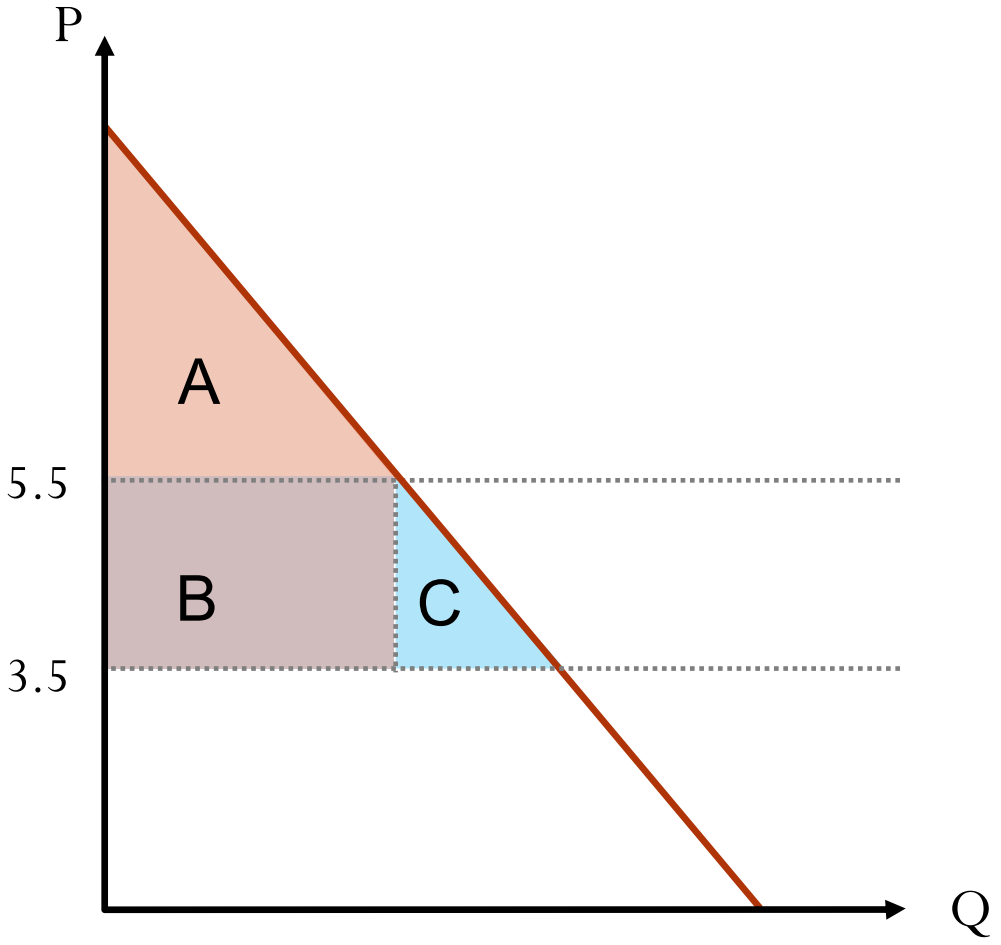
Total Consumer Surplus

The area below the demand curve and above the price line.



Comparative Statics: Price Changes

- CS before the price change: A
- CS after the price decreases: A+B+C
  - Extra surplus to existing buyers: B
  - Surplus from new buyers: C



Producer Surplus

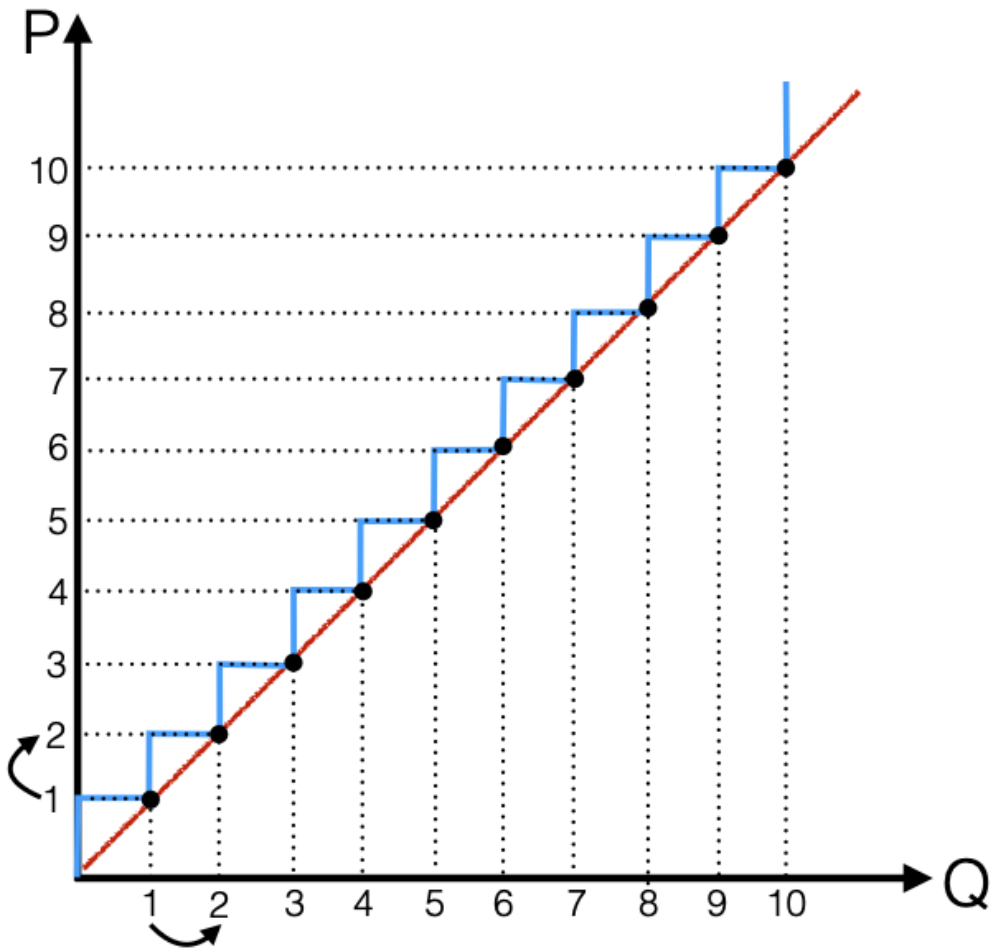
Production Cost (or Willingness to Accept)

The minimum price that a producer is willing and able to sell a good.

It measures how much the producer values everything she must give up to produce a good.

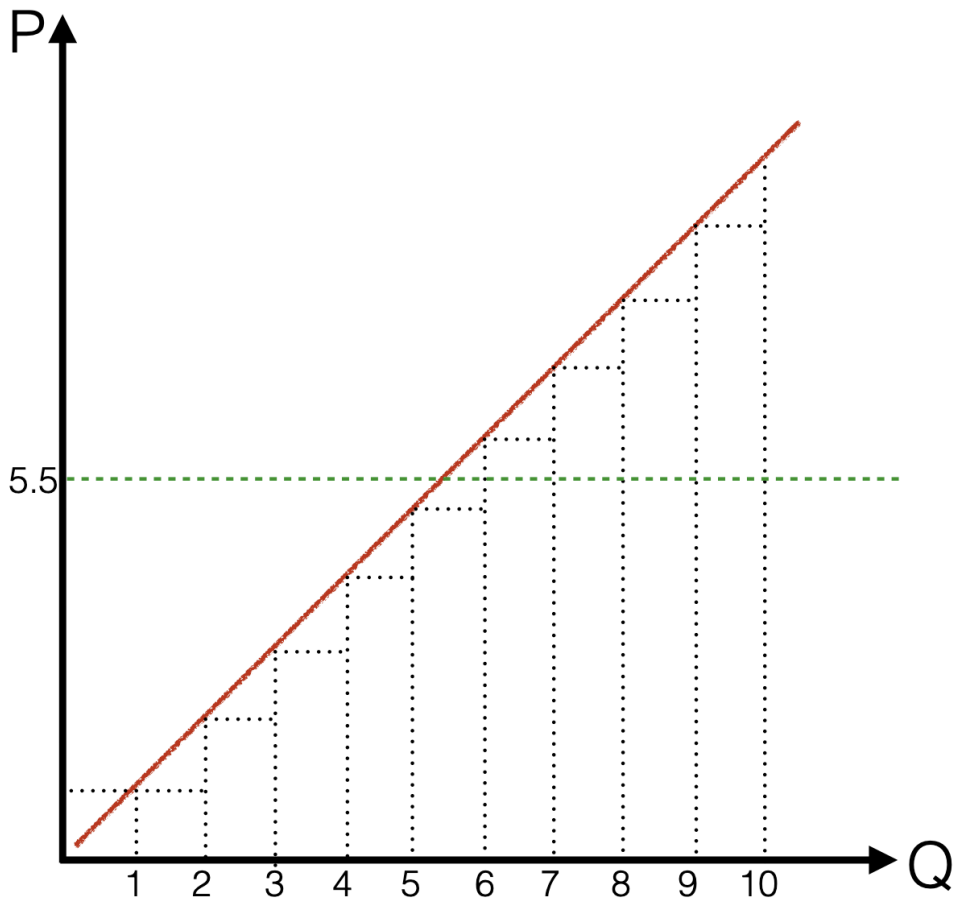
WTA, Supply, and Marginal Costs

Seller	Cost	TC	MC
1st	1	1	1
2nd	2	3	2
3rd	3	6	3
4th	4	10	4
5th	5	15	5
6th	6	21	6
7th	7	28	7
8th	8	36	8
9th	9	45	9
10th	10	55	10



Introducing Prices

Seller	Cost	Price	PS
1st	1	5.5	4.5
2nd	2	5.5	3.5
3rd	3	5.5	2.5
4th	4	5.5	1.5
5th	5	5.5	0.5
6th	6	5.5	
7th	7	5.5	
8th	8	5.5	
9th	9	5.5	
10th	10	5.5	



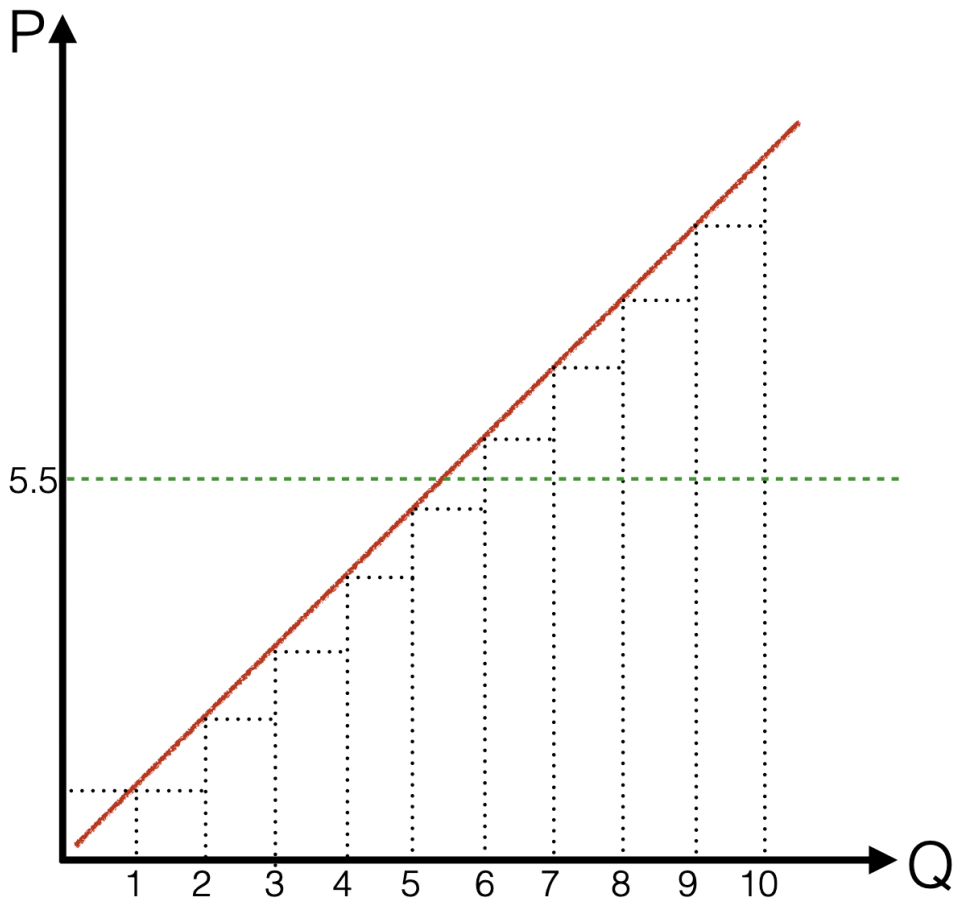
Producer Surplus

The amount that a seller is paid minus her cost of production.

It measures the benefit that sellers gain from sales.

Introducing Prices

Seller	Cost	Price	PS
1st	1	5.5	4.5
2nd	2	5.5	3.5
3rd	3	5.5	2.5
4th	4	5.5	1.5
5th	5	5.5	0.5
6th	6	5.5	0
7th	7	5.5	0
8th	8	5.5	0
9th	9	5.5	0
10th	10	5.5	0



Introducing Prices

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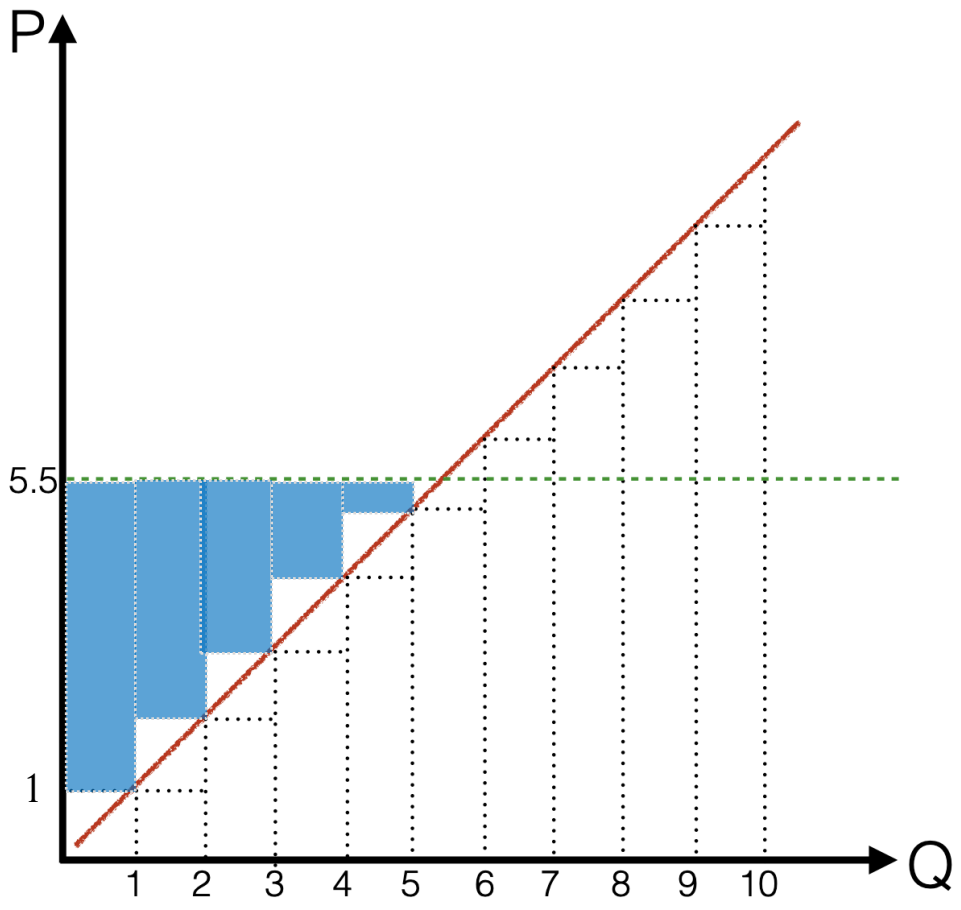
9th	9	5.5	0
10th	10	5.5	0



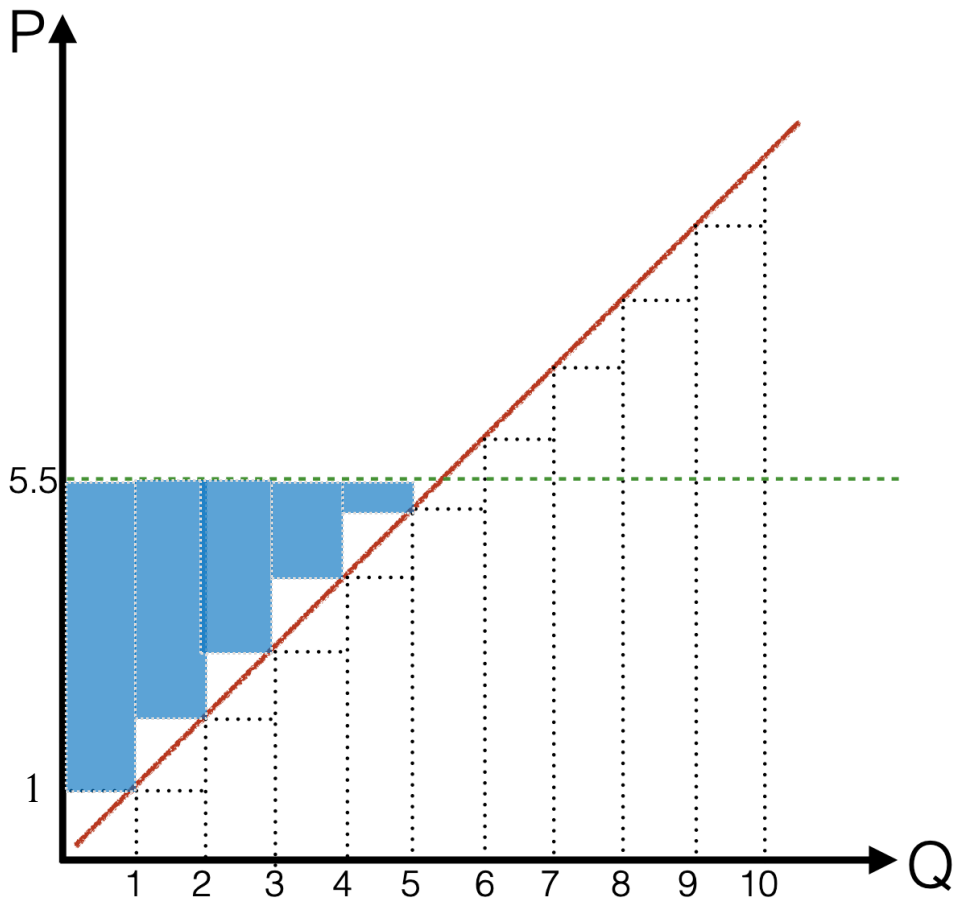
Introducing Prices

Marginal Seller's Surplus

For any given quantity, the vertical distance b/w the supply curve between price line.

Total Producer Surplus

The area above the supply curve and below the price line.



Comparative Statics: Price Changes

- PS before the price change: A
- PS after the price increases: A+B+C
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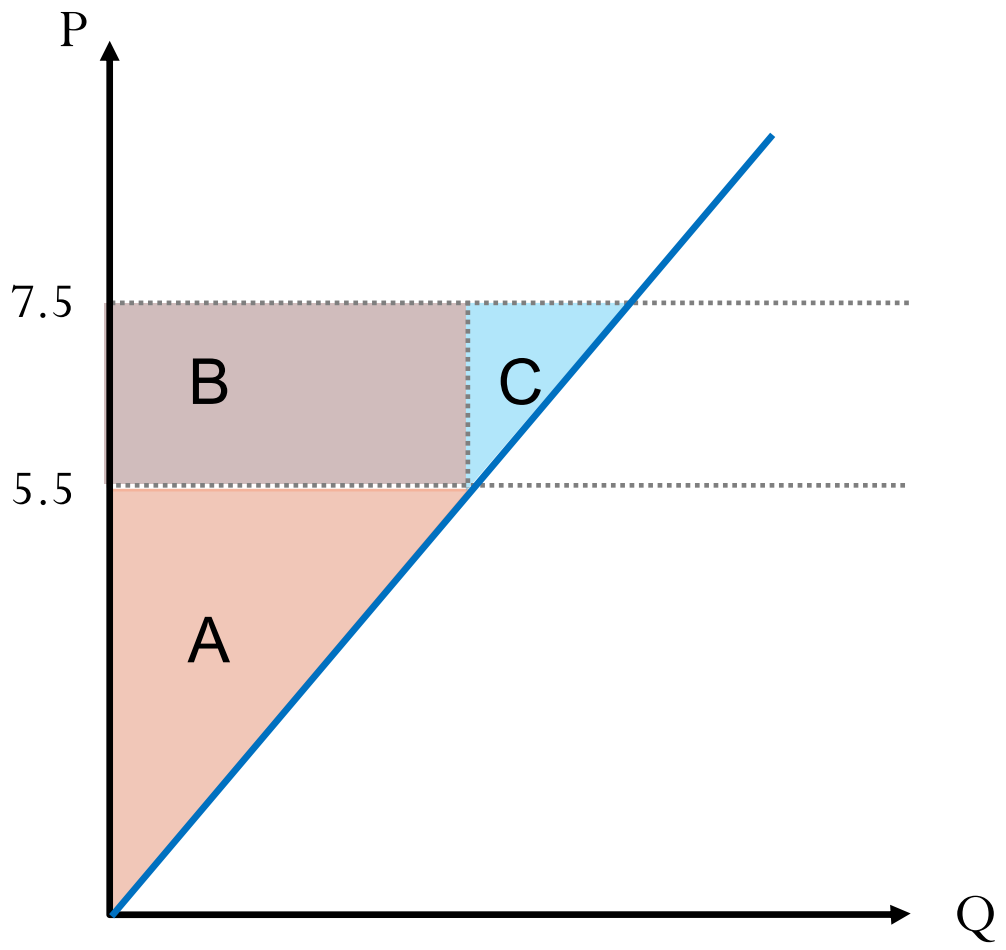
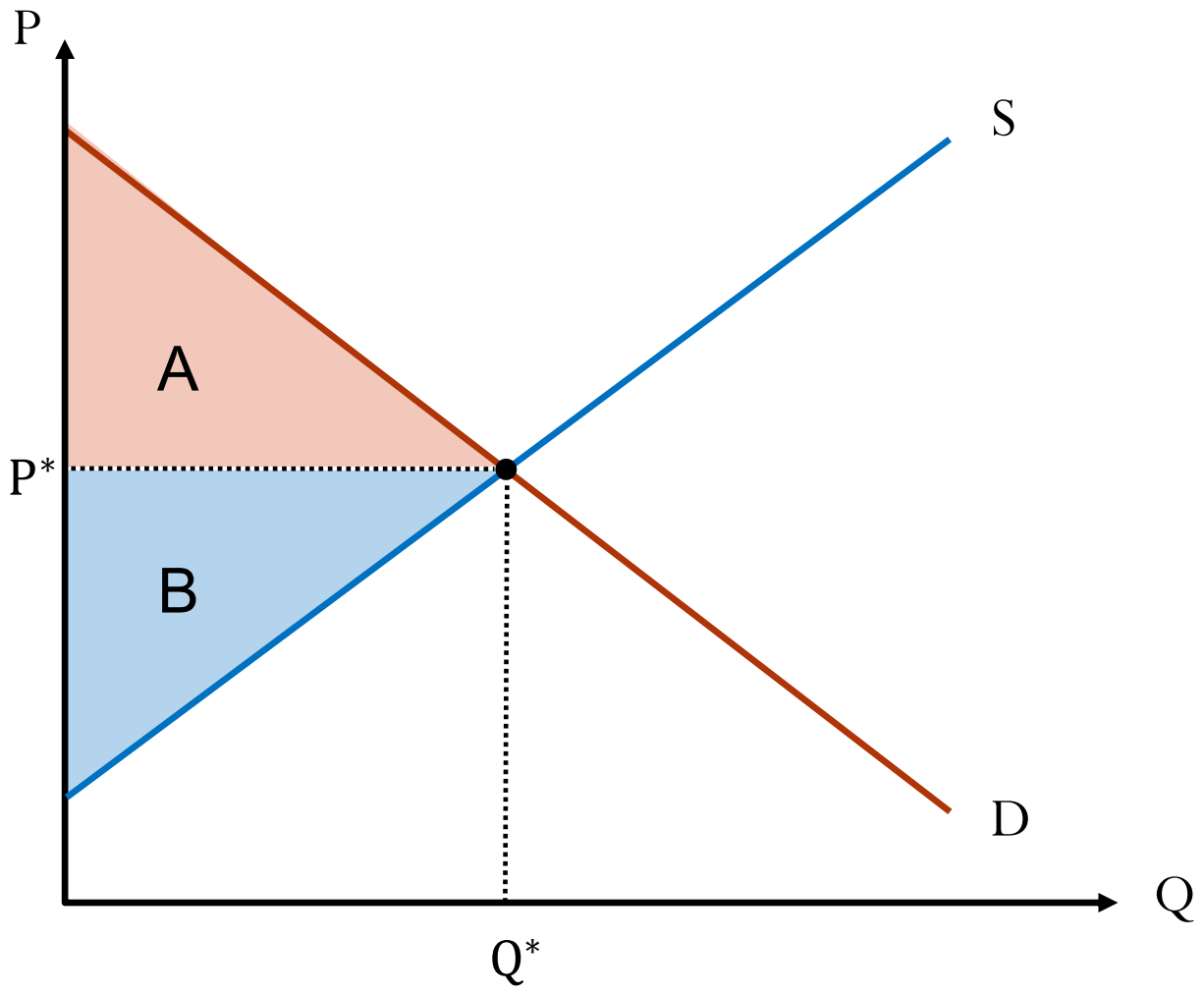


Equilibrium Surplus

Equilibrium Surplus

- Consumer Surplus: A
- Producer Surplus: B
- Total Surplus: A + B



Is the market equilibrium efficient in terms of materializing potential surplus?

Market Efficiency

Total Surplus

- Consumer surplus = Buyers' total WTP − Total price paid
- Producer surplus = Total price paid − Sellers' total production cost
- Total surplus of the society = Consumer Surplus + Producer surplus

$$\boxed{\text{Total surplus} = \text{Total WTP} - \text{Total production cost}}$$

The Benevolent Social Planner

Suppose there exists a social planner who

- knows every consumer's willingness to pay;
- knows every producer's cost of production;
- wants to maximize society's overall welfare (i.e., total surplus).

How should the planner allocate production and consumption?

The Benevolent Social Planner

Efficient (or optimal) allocation requires decisions on:

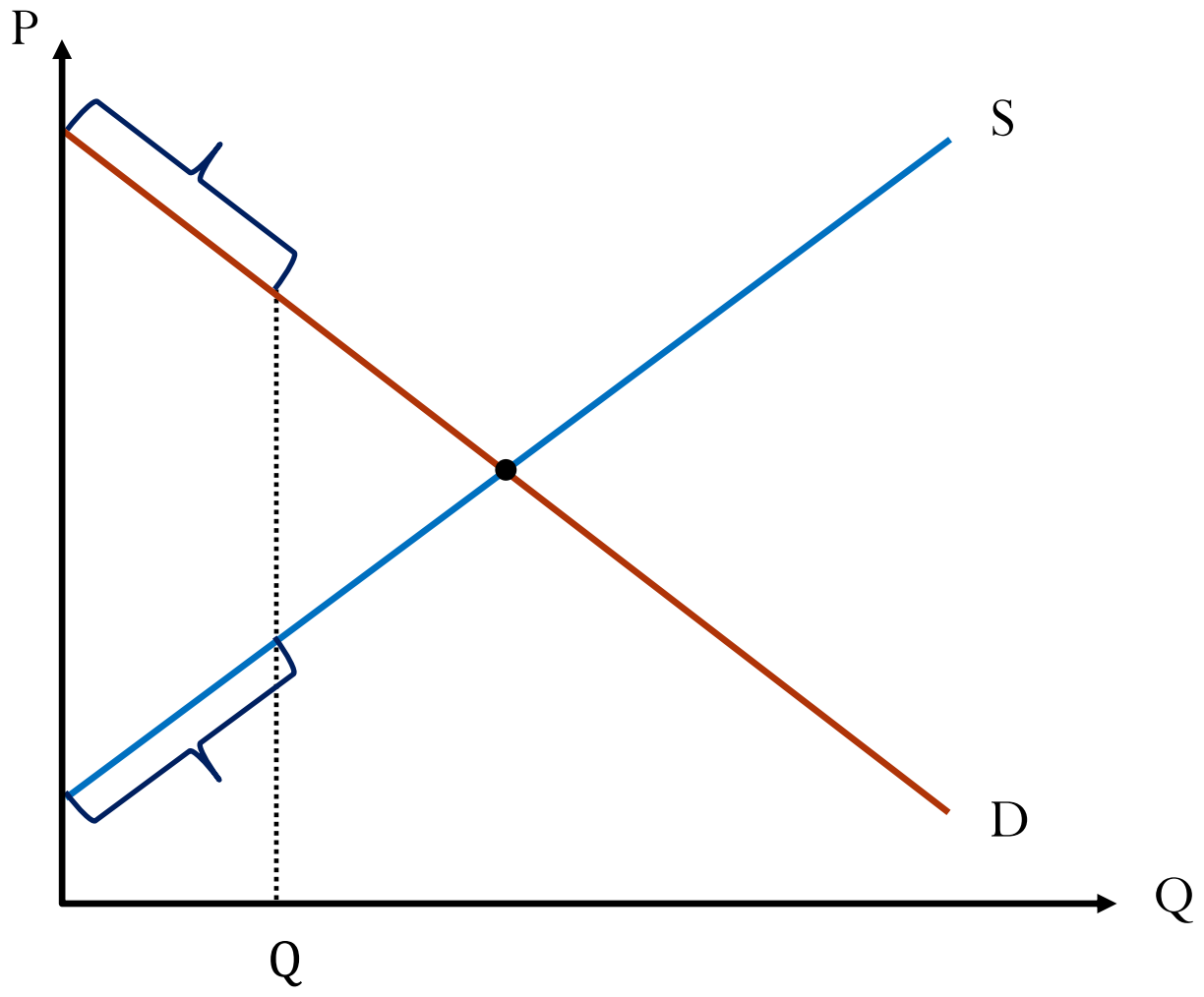
- How many units of goods should be produced?
- Which producers should produce them?
- To which consumers should the goods be distributed?

Efficient Allocation

For any given quantity of the good, a benevolent social planner should:

- let the lowest-cost producers to produce them;
- let the highest-value consumers to receive them;

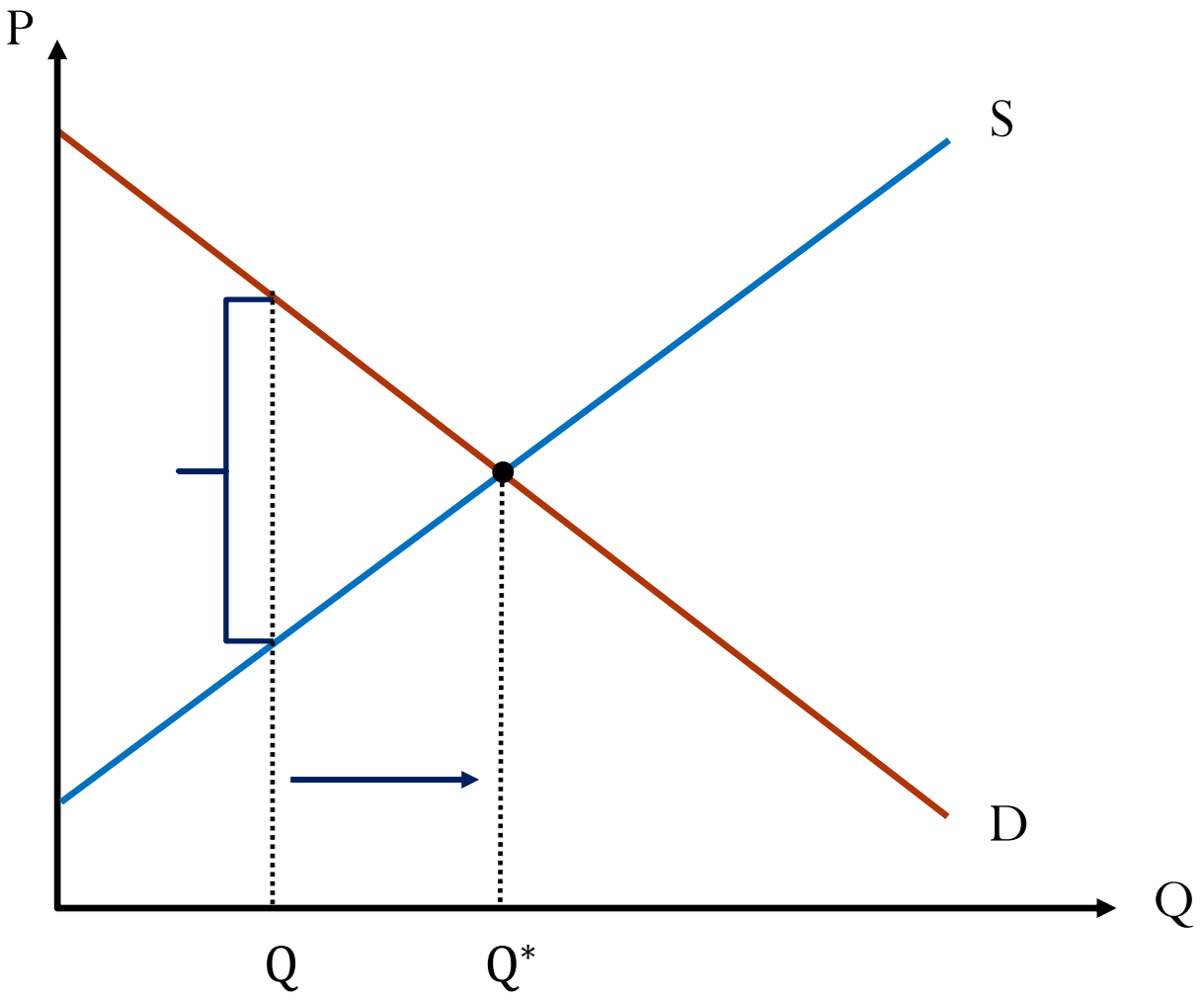
The exact matching of buyers and sellers does not affect total welfare.



Maximizing Total Surplus

When the quantity produced is lower than the equilibrium level:

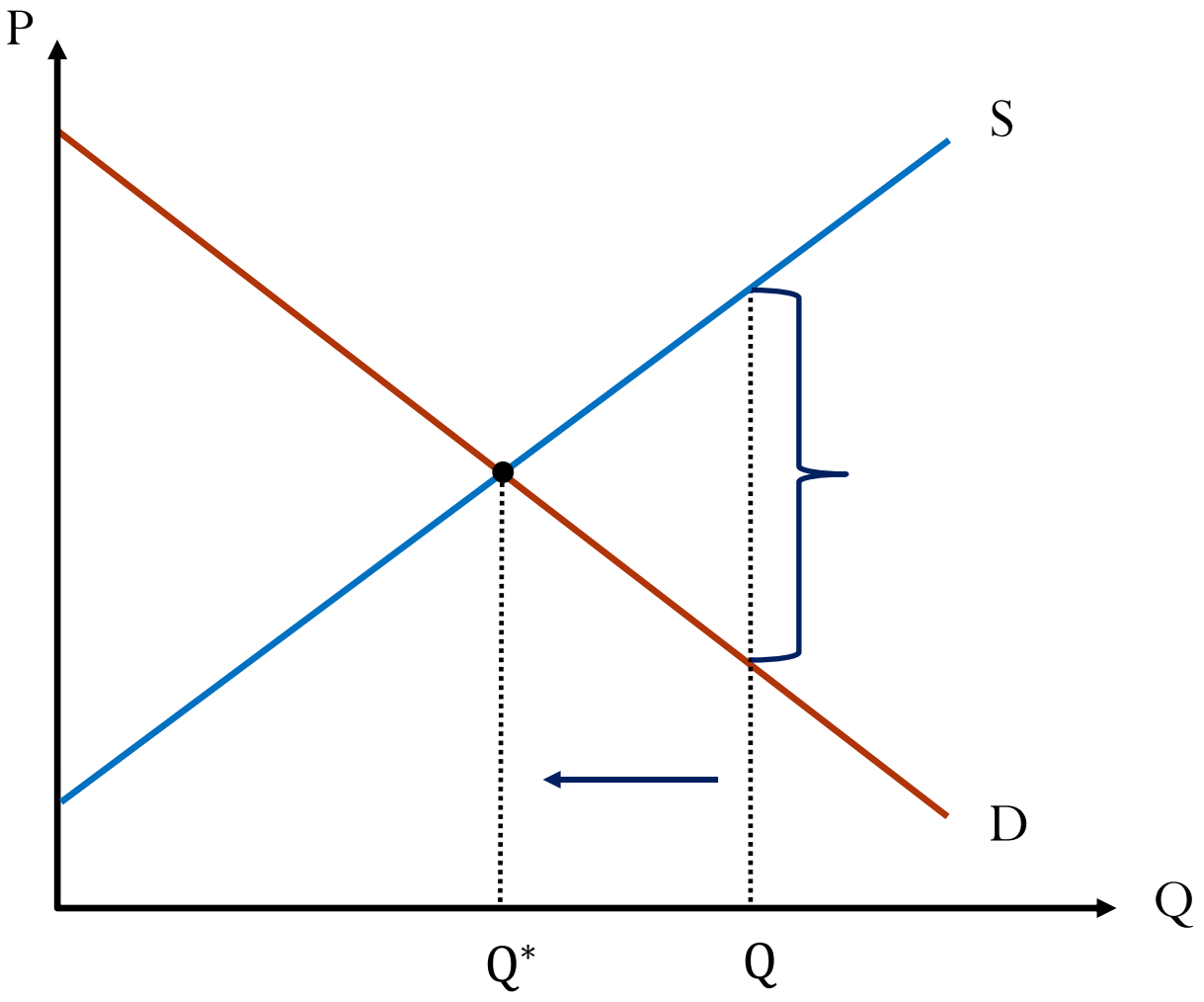
- The marginal buyer values the good more than it costs the marginal seller to make it.
- The social planner can raise total surplus by increasing output.



Maximizing Total Surplus

When the quantity produced is higher than the equilibrium level:

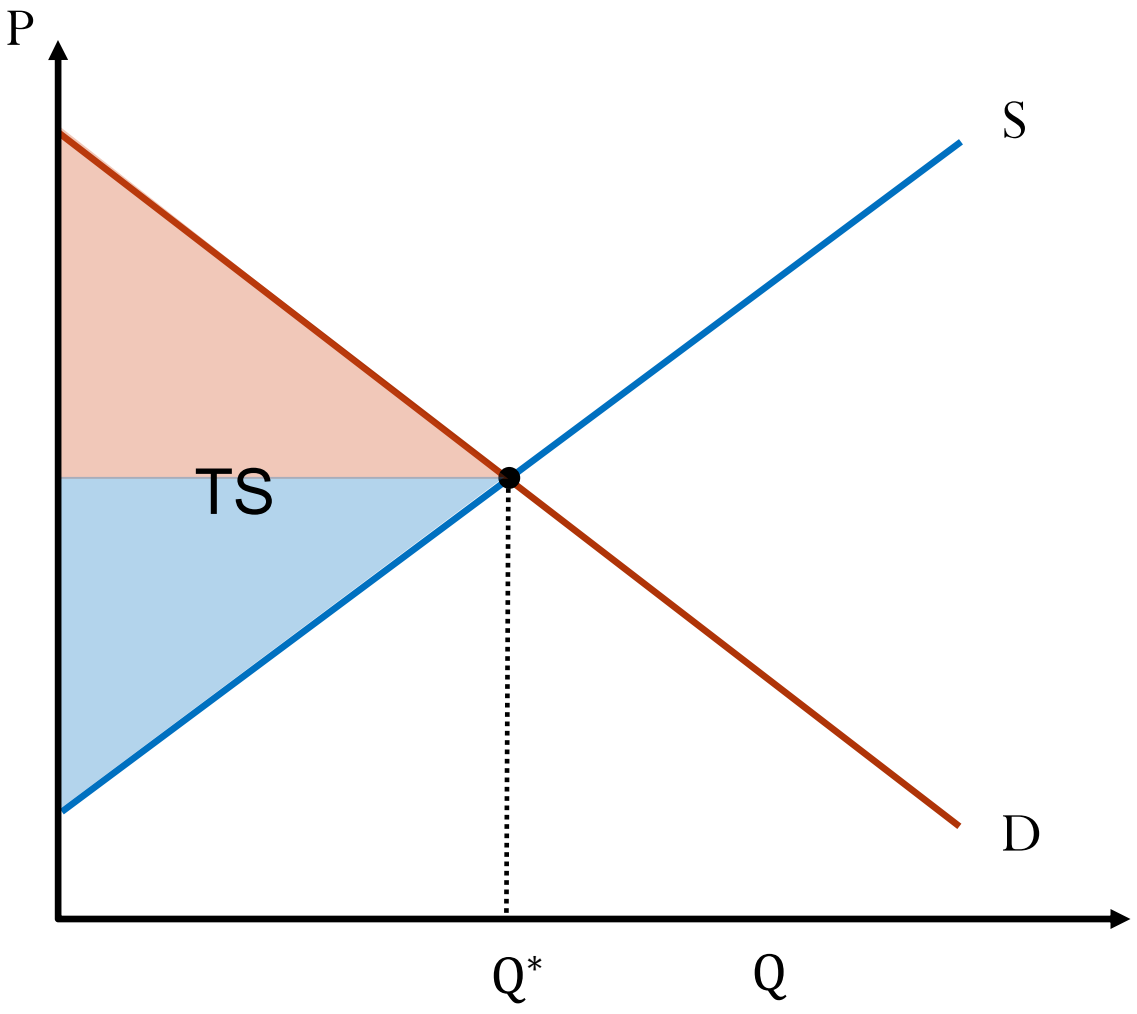
- The marginal seller cost more to produce the good than its buyer values it.
- The social planner can raise total surplus by decreasing output.



Maximizing Total Surplus

When the quantity produced equals to the equilibrium level:

- The marginal buyer's WTP equals to the the marginal seller's production cost.
- Total surplus is maximized.



Evaluating the Market Equilibrium

- Equilibrium in the market results in maximum total surplus.
- Efficient assignments: Free markets
  - assign the production of goods to the sellers who can produce them at the lowest cost;
  - assign the consumption of goods to the buyers who value them most highly.
- Efficient production:
  - Competition through the price mechanism drives the marginal buyer's WTP to equal the marginal seller's cost of production.
  - As a result, total output is guaranteed to be at the surplus-maximizing level.

Criteria for Efficiency

- Allocative Efficiency: Maximizing total social surplus.
- Pareto Efficiency: An allocation where it is impossible to make any one individual better off without making at least one individual worse off.
- What is the relationship between allocative and Pareto efficiency?
  - If an economy is allocatively efficient, is it also necessarily Pareto efficient?
  - If an economy is Pareto efficient, is it also necessarily allocatively efficient?

Criteria for Efficiency

Total social surplus is maximized $\Longleftrightarrow$ No Pareto improvement

$\Updownarrow$

Allocative Efficiency $\Longleftrightarrow$ Pareto Efficiency

Summary

- Surplus concepts: consumer, producer, and total surplus
- Market equilibrium maximizes total surplus, achieving both allocative and Pareto efficiency.
- Some 'hidden' assumptions: 1) no equity concerns; 2) perfect competition; 3) no externality.
- Next, Chapter 5: Price Controls and Quotas.